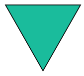
Counting the Smallest Polyiamond Containers of Every n-Iamond

The sequence a(n) counts the distinct minimum-size connected polyiamonds that contain every free `n`-iamond, up to the rotations, reflections, and translations of the triangular lattice. Each figure shows ONE representative minimal container for that `n`; the caption gives a(n) (the number of such containers) and the detail gives their common size k(n) and bounding box. Container cells are filled teal. The listed values are the number of minimum-size containers found within a finite search window whose width equals n (forced by the straight n-iamond) and whose height grows as max(4, floor((n+2)/3)); global minimality over all bounding boxes is conjectured but not separately certified.

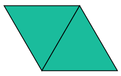
$a(1) = 1$, $a(2) = 1$, $a(3) = 1$, $a(4) = 1$, $a(5) = 4$, $a(6) = 5$, $a(7) = 39$, $a(8) = 13$, $a(9) = 26$

Computed by Peter Exley, Jun 2026

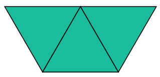
$a(1) = 1$ **[PROVED]** k=1 1x1



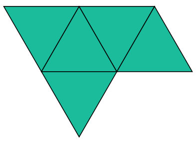
$a(2) = 1$ **[PROVED]** k=2 1x2



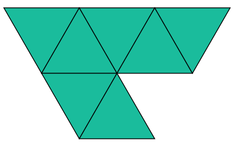
$a(3) = 1$ **[PROVED]** k=3 1x3



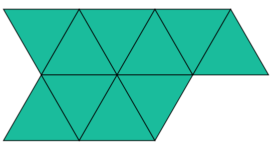
$a(4) = 1$ **[PROVED]** k=5 2x4



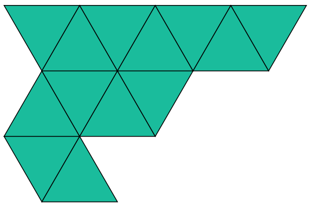
$a(5) = 4$ **[PROVED]** k=7 2x5



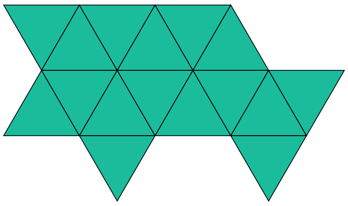
$a(6) = 5$ **[PROVED]** k=10 2x6



$a(7) = 39$ [PROVED] k=13 3x7



$a(8) = 13$ [PROVED] k=16 3x8



$a(9) = 26$ [PROVED] k=19 3x9